

Church numerals

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In lectures, the Church numerals and some associated operations are defined by

$$\begin{aligned}\underline{0} &= \lambda f. \lambda x. x \\ \underline{1} &= \lambda f. \lambda x. f x \\ \underline{2} &= \lambda f. \lambda x. f(f x) \\ &\vdots \\ \text{succ} &= \lambda a. \lambda f. \lambda x. f(a f x) \\ \text{add} &= \lambda a. \lambda b. \lambda f. \lambda x. a f(b f x)\end{aligned}$$

The following was set as an exercise:

$$\text{Show that } \underline{(n+1)} f x =_{\beta} f(\underline{n} f x).$$

I.e., show this is true for all $n \in \mathbb{N}$.

To show this, let us first define the Church numerals a little more carefully. Let us define the ‘reduced Church numerals’ $\{n\}$ to be an encoding of $\mathbb{N}$ into λ -terms t with $FV(t) \subseteq \{f, x\}$ for some fixed (distinct) variables f, x .

$$\begin{aligned}\{0\} &= x \\ \{n+1\} &= f\{n\}\end{aligned}$$

Now the usual Church numerals are given by

$$\underline{n} = \lambda f. \lambda x. \{n\}$$

for each $n \in \mathbb{N}$.

We can summarize the connections between the standard Church numerals and the reduced Church numerals by

$$\underline{n} = \lambda f. \lambda x. \{n\} \qquad \{n\} =_{\beta} \underline{n} f x$$

where the first is precise syntactic equality and the second is a β -equality.

Now we can easily verify the β -equation of the exercise for any n .

$$\begin{aligned}\underline{(n+1)} f x &=_{\beta} \{n+1\} \\ &= f\{n\} \\ &=_{\beta} f(\underline{n} f x).\end{aligned}$$